

## ADALA OMOTO MOSES

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## PROFESSIONAL SUMMARY

Results-driven **Python Backend Developer** with over 4 years of experience designing, developing, and maintaining **scalable RESTful APIs** using Django and Django Rest Framework. Strong background in **Test Driven Development (TDD)**, PostgreSQL optimization, and secure system design. Proven ability to translate real-world academic and business requirements into reliable backend solutions. Experienced ICT lecturer with a strong foundation in cybersecurity and system architecture. Seeking remote backend or software engineering roles.

## TECHNICAL SKILLS

### Languages:

Python, JavaScript, C

### Backend & APIs:

Django, Django Rest Framework, Flask, RESTful API Design, JWT Authentication

### Databases:

PostgreSQL, MySQL, MongoDB, SQL Optimization, Full-Text Search

### Dev & Tools:

Linux (Ubuntu), Git, GitHub, Docker (basic), Stripe API

### Testing & Quality:

Test Driven Development (TDD), Pytest, Django TestCase

### Other:

Machine Learning Algorithms, Deep Learning Fundamentals, System Security

## PROJECTS

### EduCore – Scalable LMS Backend API

#### Django | DRF | PostgreSQL | JWT | TDD

- Designed and developed a **production-ready Learning Management System backend API** supporting Admin, Lecturer, and Student roles.
- Implemented **JWT-based authentication** and **role-based access control** to secure academic workflows.
- Built course management, enrollment, and assignment modules using Django Rest Framework.
- Applied **Test Driven Development (TDD)** to ensure reliability, security, and maintainability.
- Modeled real-world academic processes into scalable backend architecture.

**GitHub:** [https://github.com/mosese-maker/ EduCore-LMS-Backend-API-Django-DRF-](https://github.com/mosese-maker/EduCore-LMS-Backend-API-Django-DRF-)

### SmartSearch – PostgreSQL Full-Text Search API

#### Django | PostgreSQL | SQL Optimization

- Built a high-performance REST API implementing **PostgreSQL full-text search** using `tsvector` and `GIN` indexes.
- Implemented relevance ranking, pagination, and optimized query performance.
- Achieved up to **60% faster search response times** through database indexing and query tuning.
- Demonstrated advanced SQL and database optimization techniques.

**GitHub:** <https://github.com/mosese-maker/smartsearch-api>

### SecureShop – E-Commerce Backend API

#### Django | PostgreSQL | Stripe

- Developed a secure e-commerce backend with product catalog, cart, and order management.
- Integrated **Stripe payment gateway** with webhook verification.
- Improved transaction processing efficiency and data integrity.

## PROFESSIONAL EXPERIENCE

### East Africa Institute of Certified Studies — *Lecturer (ICT & Engineering)*

September 2022 – Present

- Supervised and maintained ICT laboratories, reducing repair costs by **50%** through preventive maintenance.
- Delivered hands-on training in **Cybersecurity, Computer Maintenance, and Software Tools**, achieving **90% student retention**.
- Achieved **95% student satisfaction** through practical, project-based teaching methods.
- Integrated technology-driven learning tools, improving student engagement by **15%**.
- Taught video editing and digital content tools, with **80% of students scoring above average**.

### Python Developer — *Personal & Client Projects*

2020 – Present

- Designed and maintained scalable **RESTful APIs** using Django and Flask.
- Applied **TDD principles** to ensure code quality and reliability.
- Implemented PostgreSQL **full-text search**, improving data retrieval efficiency by **60%**.
- Developed backend systems for e-commerce, academic, and data-driven applications.

## EDUCATION

### Bachelor of Science – Computer Security & Forensics

Jaramogi Oginga Odinga University of Science and Technology

## CERTIFICATIONS

- Python Django Developer – LinkedIn Learning
- Python Essentials – Cisco Networking Academy
- Full Stack Web Development with Flask – LinkedIn Learning
- Python Developer – Sololearn
- Database Administrator – MongoDB University
- Node.js Developer Track – MongoDB University
- SQL (Basic & Intermediate) – HackerRank
- REST API (Intermediate) – HackerRank
- React (Basic) – HackerRank

- Deep Learning Using Python – Udacity
- Scientific Computing with Python – freeCodeCamp
- Open-Source Masterclass – Certificate of Completion

## **AVAILABILITY**

- Open to **remote and contract opportunities**
- Comfortable working with distributed teams across time zones